

SAFETY DATA SHEET

according to the Globally Harmonized System



CATULIA™

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	02.12.2025	50002791	Date of first issue: 02.12.2025

1. IDENTIFICATION

Product name : CATULIA™

Manufacturer or supplier's details

Company : FMC CORPORATION

Address : 2929 WALNUT STREET
PHILADELPHIA, PA 19104 USA
(215) 299-6000 (GENERAL INFORMATION)

E-mail address : SDS-Info@fmc.com

Emergency telephone : 1 703 / 741-5970 (CHEMTREC - International)

Medical Emergency Number : CALL 800-10-6966, JAPANESE UNIVERSITY HOSPITAL
POISON INFORMATION CENTER. SANTA CRUZ-BOLIVIA.

Recommended use of the chemical and restrictions on use

Recommended use : Biological fungicide/nematicide

Restrictions on use : Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Not a hazardous substance or mixture.

GHS label elements

No hazard pictogram, no signal word, no hazard statement(s), no precautionary statement(s) required.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
sodium hydrogensulphate	7681-38-1	$\geq 1 - < 3$

4. FIRST AID MEASURES

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General advice	: Do not leave the victim unattended.
If inhaled	: If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a physician.
In case of skin contact	: Wash off with soap and water. If symptoms persist, call a physician. Wash contaminated clothing before re-use.
In case of eye contact	: Remove contact lenses. Protect unharmed eye. If eye irritation persists, consult a specialist.
If swallowed	: Keep respiratory tract clear. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.
Most important symptoms and effects, both acute and delayed	: None known.
Protection of first-aiders	: Avoid inhalation, ingestion and contact with skin and eyes.
Notes to physician	: Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	: Dry chemical, CO2, water spray or regular foam.
Unsuitable extinguishing media	: High volume water jet Do not spread spilled material with high-pressure water streams.
Hazardous combustion products	: Carbon oxides Ammonia Sulfur oxides phosphorus oxides
Specific extinguishing methods	: Remove undamaged containers from fire area if it is safe to do so. Use a water spray to cool fully closed containers. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment	: Firefighters should wear protective clothing and self-contained

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for fire-fighters

breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

- | | |
|---|---|
| Personal precautions, protective equipment and emergency procedures | : Evacuate personnel to safe areas.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Avoid formation of aerosol. |
| Environmental precautions | : If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and materials for containment and cleaning up | : Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal. |

7. HANDLING AND STORAGE

- | | |
|---|---|
| Advice on protection against fire and explosion | : Normal measures for preventive fire protection. |
| Advice on safe handling | : For personal protection see section 8.
Avoid formation of respirable particles.
Dispose of rinse water in accordance with local and national regulations.
Smoking, eating and drinking should be prohibited in the application area. |
| Conditions for safe storage | : Electrical installations / working materials must comply with the technological safety standards. |
| Materials to avoid | : Strong oxidizing agents
Strong acids and strong bases |
| Further information on storage stability | : No decomposition if stored and applied as directed. |

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection	: In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.
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Hand protection

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Material	: Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
Eye protection	: Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	: Choose body protection according to the amount and concentration of the dangerous substance at the work place. Protective suit
Protective measures	: Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions.
Hygiene measures	: Avoid contact with skin, eyes and clothing. Do not inhale aerosol. When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: liquid
Color	: brown
Odor	: Bitter, musty
Odor Threshold	: No data available
pH	: 4,79 Method: OCSP 830.7000 (at 1% suspension) 4,12 (undiluted)
Melting point/freezing point	: not determined
Boiling point/boiling range	: not determined
Flash point	: Not applicable
Evaporation rate	: No data available
Flammability (liquids)	: Not expected to be ignitable

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Self-ignition	:	not determined
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	1,21 (20 °C)
Density	:	1,21 g/cm ³ (20 °C)
Solubility(ies) Water solubility	:	suspendable
Partition coefficient: n-octanol/water	:	Not applicable
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity Viscosity, dynamic	:	1.154,9 mPa.s (40 °C) 30 rpm 598,9 mPa.s (40 °C) 100 rpm 1.443,6 mPa.s (20 °C) 30 rpm 743,9 mPa.s (20 °C) 100 rpm
Viscosity, kinematic	:	not determined
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Surface tension	:	35 mN/m, 20 °C, OECD Test Guideline 115
Molecular weight	:	Not applicable

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10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	Stable under recommended storage conditions.
Conditions to avoid	:	Avoid extreme temperatures. Avoid formation of aerosol. Protect from frost, heat and sunlight.
Incompatible materials	:	Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	:	Stable under recommended storage conditions.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Based on available data, the classification criteria are not met.

Product:

Acute oral toxicity	:	LD50(Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 425 Assessment: The substance or mixture has no acute oral toxicity LD50(Rat, male and female): >3.1x10 ⁹ MCA/animal Method: OPPTS 885.3050 Assessment: The substance or mixture has no acute oral toxicity
Acute inhalation toxicity	:	LC50(Rat, male and female): > 5,45 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	:	LD50(Rat, male and female): > 5.000 mg/kg Method: OECD Test Guideline 402

Components:

sodium hydrogensulphate:

Acute oral toxicity	:	LD50 (Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 423 Remarks: Based on data from similar materials
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Acute inhalation toxicity : LC0 (Rat, male and female): > 2,4 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 436
Remarks: Based on data from similar materials
no mortality

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
Remarks : Minimal effects that do not meet the threshold for classification.
Based on data from a similar product.

Components:

sodium hydrogensulphate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation
Remarks : Minimal effects that do not meet the threshold for classification.
Based on data from a similar product.

Components:

sodium hydrogensulphate:

Species : Rabbit
Method : OECD Test Guideline 405
Result : Irreversible effects on the eye

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

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Product:

Test Type	: Local lymph node assay (LLNA)
Routes of exposure	: Skin contact
Species	: mice
Method	: OECD Test Guideline 429
Result	: Not a skin sensitizer.

Components:

sodium hydrogensulphate:

Test Type	: Maximization Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Does not cause skin sensitization.
Remarks	: Based on data from similar materials

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

sodium hydrogensulphate:

Genotoxicity in vitro	: Test Type: reverse mutation assay Test system: TA100 Result: negative Remarks: Based on data from similar materials
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	Test Type: gene mutation test Test system: mouse lymphoma cells Method: OECD Test Guideline 476 Result: negative Remarks: Based on data from similar materials
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	Test Type: Chromosome aberration test in vitro Test system: Chinese hamster fibroblasts Method: OECD Test Guideline 473 Remarks: Based on data from similar materials
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Germ cell mutagenicity - Assessment	: In vitro tests did not show mutagenic effects
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Carcinogenicity

Based on available data, the classification criteria are not met.

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

sodium hydrogensulphate:

Effects on fertility	: Species: Rat, male and female Application Route: Oral
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Dose: 100, 300, 1000 mg/kg bw/day
General Toxicity Parent: NOEL: 1.000 mg/kg body weight
Method: OECD Test Guideline 421
Result: negative
Remarks: Based on data from similar materials

Effects on fetal development : Test Type: reproductive and developmental toxicity study
Species: Mouse
Application Route: Oral
Dose: 2800 mg/kg/day
General Toxicity Maternal: NOAEL: 2.800 mg/kg body weight
Developmental Toxicity: NOAEL: 2.800 mg/kg body weight
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

Based on available data, the classification criteria are not met.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:

sodium hydrogensulphate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

sodium hydrogensulphate:

Species : Rat, male and female
NOAEL : 1.000 mg/kg
Application Route : Oral
Exposure time : 7 weeks
Dose : 100, 300, 1000 mg/kg bw/day
Method : OECD Test Guideline 421
Remarks : Based on data from similar materials

Species : Rabbit, male and female
LOAEL : 2 mL/kg/day(16% w/w)
Application Route : Dermal
Exposure time : 91 d
Dose : 2 ml/kg/day-16 % w/w aq.- Sodi
Method : OECD Test Guideline 411
Remarks : Based on data from similar materials

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Aspiration toxicity

Based on available data, the classification criteria are not met.

Further information

Product:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish : LD50 (Oncorhynchus mykiss (rainbow trout)): , $>1.4 \times 10^6$ cfu/ml
Exposure time: 30 d
Test Type: semi-static test
Method: US EPA Test Guideline OPPTS 885.4200
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EC50: $>1 \times 10^6$ CFU/ml
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Test Type: Reproduction Test
Method: OECD Test Guideline 211
Remarks: Based on data from similar materials

NOEC: 1×10^6 CFU/ml
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Test Type: Reproduction Test
Method: OECD Test Guideline 211
Remarks: Based on data from similar materials

Toxicity to soil dwelling organisms : EC50: $>4 \times 10^9$ CFU/kg artificial soil
Exposure time: 28 d
End point: reproduction
Species: Folsomia candida
Method: OPPTS 885.4340
Remarks: Based on data from similar materials

NOEC: $\geq 4 \times 10^9$ CFU/kg artificial soil
Exposure time: 28 d
End point: reproduction
Species: Folsomia candida
Method: OPPTS 885.4340
Remarks: Based on data from similar materials

NOEC: $\geq 4 \times 10^9$ CFU/kg artificial soil
Exposure time: 14 d
End point: reproduction
Species: Hypoaspis aculeifer

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Method: OPPTS 885.4340
Remarks: Based on data from similar materials

EC50: $>4 \times 10^9$ CFU/kg artificial soil
Exposure time: 14 d
End point: reproduction
Species: *Hypoaspis aculeifer*
Method: OPPTS 885.4340
Remarks: Based on data from similar materials

Components:

sodium hydrogensulphate:

Toxicity to fish : LC50 (*Pimephales promelas* (fathead minnow)): 7.960 mg/l
Exposure time: 96 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : LC50 (*Daphnia magna* (Water flea)): 1.766 mg/l
Exposure time: 48 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (Marine Diatom): 1.900 mg/l
Exposure time: 120 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to microorganisms : NOEC (activated sludge): 8 g/l
Exposure time: 37 d
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : LOEC: 1.329 mg/l
Exposure time: 7 d
Species: *Ceriodaphnia dubia* (water flea)
Test Type: semi-static test
Remarks: Based on data from similar materials

Persistence and degradability

No data available

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

sodium hydrogensulphate:

Bioaccumulation : Bioconcentration factor (BCF): 0,5
Method: QSAR
Remarks: Bioaccumulation is unlikely.

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Mobility in soil

No data available

Other adverse effects

Product:

Additional ecological information : No data available

13. DISPOSAL CONSIDERATIONS

Disposal methods

- Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.
- Contaminated packaging : Empty remaining contents.
Packaging that is not properly emptied must be disposed of as the unused product.
Do not re-use empty containers.
Empty containers should be taken to an approved waste handling site for recycling or disposal.

It is prohibited to reuse, bury, burn, or sell containers. Rinsable containers: Triple rinse containers of less than 20 liters and pressure rinse containers of 20 liters or more. Triple rinsing: Add water up to $\frac{1}{4}$ of the container's capacity, close and shake for 30 seconds. Pour the rinse water into the mixing tank, considering this volume of water within the recommended volume for mixing preparation. Perform this procedure three times. Pressure rinsing: Activate the pressure rinsing device for 30 seconds, considering the volume of water used as part of the recommended volume for mixing preparation. In both procedures, punctured the container on its base without damaging the label. In all cases, take the empty containers to collection points indicated by the local empty containers program.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

Not regulated as a dangerous good

IATA-DGR

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Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to IMO instruments

Not applicable for product as supplied.

Special precautions for user

Remarks : Not classified as dangerous in the meaning of transport regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The ingredients of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. Bacillus subtilis strain FMCH002 Bacillus Paralicheniformis strain FMCH001
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

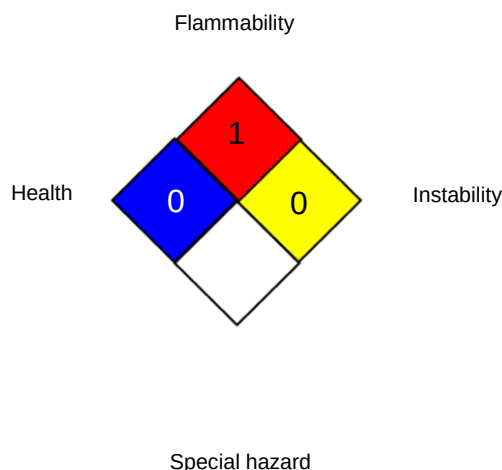
Revision Date	: 02.12.2025
Date format	: dd.mm.yyyy

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Further information

NFPA:



HMIS® IV:

HEALTH	/	0
FLAMMABILITY		1
PHYSICAL HAZARD		0

HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recom-

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recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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